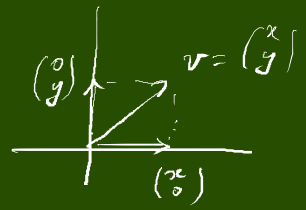
$\langle u, v \rangle = u^t \cdot v \qquad u, v \in \mathbb{R}^n$

$\mathbb{R}^2$



$u, v \neq 0 \qquad \left| \underbrace{\frac{\langle u, v \rangle}{\|u\| \cdot \|v\|}}_{\cos\theta} \right| \leq 1$

$\theta \in [0, \pi]$

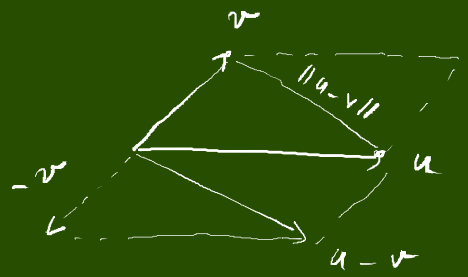
$\cos\theta = \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|}$

$\theta = 0 \qquad \cos\theta = 1 \qquad \langle u, v \rangle = \|u\| \cdot \|v\|$
$\iff v = \lambda u, \quad \lambda > 0$

$\theta = \pi \qquad \cos\theta = -1 \qquad \langle u, v \rangle = -\|u\| \cdot \|v\|$
$\iff v = \lambda u, \quad \lambda < 0$

$\theta = \frac{\pi}{2} \qquad \cos\theta = 0 \qquad \langle u, v \rangle = 0$

u, v ortogonais $u \perp v$



$(X, d : X \times X \longrightarrow \mathbb{R}_+)$ sp. métrico

distância (métrica)

$x = 110101$

$y = 100011$

HAMMING

$d(x,y) = 3$

$x = 110101$

$z = 10011$

LEVENSHTEIN

delete
insert
replace

$d(x,z) \leq 3$

$d(x,z) = 2$ V



$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

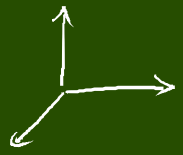
$$\langle e_i, e_j \rangle = 0, \quad i \neq j$$

$$\|e_i\| = 1 = \sqrt{\langle e_i, e_i \rangle} \Rightarrow \langle e_i, e_i \rangle = 1$$

$$\langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

Kronecker

$$V = \{ f : [0, 2\pi] \to \mathbb{R} \mid f \text{ cont.} \}$$



$$\langle f, g \rangle = \int_0^{2\pi} f(t) g(t) \, dt$$

$\Rightarrow$ $1 = \cos 0x$, $\cos x, \sin x, \cos 2x, \sin 2x, \ldots, \cos nx, \sin nx$

ortog.

$\langle \cos nx, \cos mx \rangle = 0$, $n \neq m$

$\langle \cos nx, \sin mx \rangle = 0$ $\forall n, m$

$\langle \sin mx, \sin nx \rangle = 0$ $\forall n \neq m$.

$\longrightarrow$ FOURIER MS

$(V, \langle , \rangle)$ $v_1 \ldots v_n \in V$, $v_1 \ldots v_n \neq 0$

ortogonali

$\Rightarrow$ $v_1 \ldots v_n$ liniar indep.

Fie $\lambda_1 \ldots \lambda_n \in \mathbb{R}$ a.i. $\lambda_1 v_1 + \ldots + \lambda_n v_n = 0 \;/ \cdot \langle v_1, - \rangle$

$\lambda_1 \underbrace{\langle v_1, v_1 \rangle}_{\neq 0} + \lambda_2 \underbrace{\langle v_1, v_2 \rangle}_{0} + \ldots + \lambda_n \underbrace{\langle v_1, v_n \rangle}_{0} = 0 \Rightarrow \lambda_1 = 0$

analog $\lambda_2 = \ldots = \lambda_n = 0$.

$\lambda_1 \cdot 1 + \lambda_2 \cdot \cos x + \lambda_3 \cdot \sin x + \lambda_4 \cos 2x + \ldots = 0$

$\int_0^{2\pi} (\text{---}) \cdot 1 \, dx = 0$

$= \lambda_1 \cdot 2\pi$

$\Rightarrow \lambda_1 = 0$

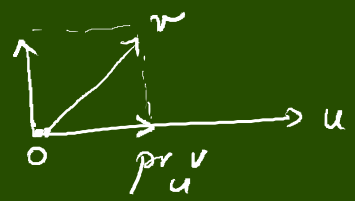
$\int_0^{2\pi} (\text{---}) \cos x \, dx = 0$

$= \lambda_2 \cdot \underbrace{\int_0^{2\pi} \cos^2 x \, dx}_{\neq 0}$

$\Rightarrow \lambda_2 = 0$ etc.

$pr_u v$

$u \neq 0$



unicitatea

$v = w_1 + w_2 \quad \begin{cases} w_1 \in \mathcal{L}\{u\} \\ w_2 \perp u \end{cases}$

$v = z_1 + z_2 \quad \begin{cases} z_1 \in \mathcal{L}\{u\} \\ z_2 \perp u \end{cases}$

$\Rightarrow \underbrace{w_1 - z_1}_{\in \mathcal{L}\{u\}} = \underbrace{z_2 - w_2}_{}$

$\langle z_2 - w_2, z_2 - w_2 \rangle = \langle w_1 - z_1, z_2 - w_2 \rangle =$

$= \underbrace{\langle w_1, z_2 \rangle}_{\perp u} - \underbrace{\langle w_1, w_2 \rangle}_{0} - \underbrace{\langle z_1, z_2 \rangle}_{0} + \underbrace{\langle z_1, w_2 \rangle}_{0}$

$W_1 = \alpha u$

$\langle w_1, z_2 \rangle = \langle \alpha u, z_2 \rangle = \alpha \langle u, z_2 \rangle = 0$

$\Rightarrow \quad \langle z_2 - w_2, z_2 - w_1 \rangle = 0 \Rightarrow z_2 - w_2 = 0$

$\Rightarrow z_2 = w_2 \Rightarrow z_1 = w_1$

existența $\quad V = W_1 + W_2 \qquad W_1 = \alpha u$

$W_2 \perp u \;/\Rightarrow$

$\Rightarrow W_2 = V - \alpha u$

$\Rightarrow 0 = \langle w_2, u \rangle = \langle v - \alpha u, u \rangle =$

$= \langle v, u \rangle - \alpha \langle u, u \rangle$

$\Rightarrow \alpha = \dfrac{\langle v, u \rangle}{\langle u, u \rangle} \qquad \langle u, u \rangle \neq 0 \quad (u \neq 0)$

$\Rightarrow v = \underbrace{\dfrac{\langle v, u \rangle}{\langle u, u \rangle} u}_{\in Sp\{u\}} + \underbrace{\left(v - \dfrac{\langle v, u \rangle}{\langle u, u \rangle} u \right)}_{\perp u}$

pr_u^v

teorema $\|\text{pr}_u v\| \leq \|v\| \quad \forall v, u \neq 0$

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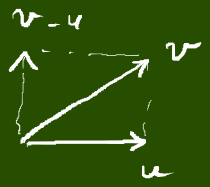
ex. $u = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \qquad v = \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix}$

$u \perp v$? $\quad \langle u, v \rangle = 2 \cdot 6 + 1 \cdot (-3) + 2 \cdot 0 = 9 \neq 0$

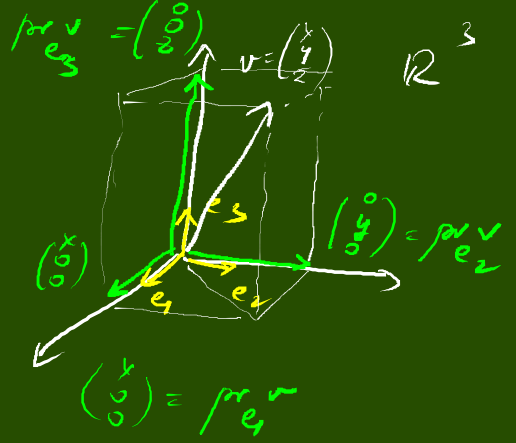
$u \not\perp v$

$\text{pr}_u v = ?$ $\quad$ (obs: $u \perp v \Rightarrow \text{pr}_u v = 0$)

$$\text{pr}_u v = \frac{\langle u, v \rangle}{\langle u, u \rangle} u = \frac{9}{9} u = u$$



$v - u = \begin{pmatrix} 4 \\ -4 \\ -2 \end{pmatrix} \perp u$



V $\{v_1 \dots v_n\}$ <u>ortogonalė</u>

$v \in V \Rightarrow \exists!\ \lambda_1 \dots \lambda_n \in \mathbb{R}$

$v = \lambda_1 v_1 + \dots + \lambda_n v_n \quad / \cdot \langle v_1, - \rangle$

$\langle v_1, v \rangle = \lambda_1 \underbrace{\langle v_1, v_1 \rangle}_{\neq 0} + \lambda_2 \underbrace{\langle v_1, v_2 \rangle}_{0} + \dots + \lambda_n \underbrace{\langle v_1, v_n \rangle}_{0}$

$\Rightarrow \lambda_1 = \dfrac{\langle v_1, v \rangle}{\langle v_1, v_1 \rangle}$ analogiškai $\lambda_i = \dfrac{\langle v_i, v \rangle}{\langle v_i, v_i \rangle}$

$$[u]_B = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \iff u = x_1 v_1 + \dots + x_n v_n$$

$$[v]_B = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \iff v = y_1 v_1 + \dots + y_n v_n$$

$$\langle u, v \rangle = \langle x_1 v_1 + \dots + x_n v_n, \; y_1 v_1 + \dots + y_n v_n \rangle =$$

$$= \sum_{i,j=1}^{n} x_i y_j \underbrace{\langle v_i, v_j \rangle}_{\begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}} = \underbrace{x_1 y_1 + \dots + x_n y_n}_{\mathbb{R}^n}$$

$\{v_1, \dots v_n\}$ — baza arbitrară

$\longrightarrow \{u_1, \dots u_n\}$ — bază ortogonală

$\longrightarrow \{e_1, \dots e_n\}$ — bază ortonormată

$v \neq 0 \implies w \in Sp\{v\}$ $\quad \|w\| = 1$?



$w = \alpha v$

$\|w\| = 1 \quad |\alpha| \cdot \|v\| = 1$

$|\alpha| = \frac{1}{\|v\|} \qquad \alpha = \pm \frac{1}{\|v\|}$

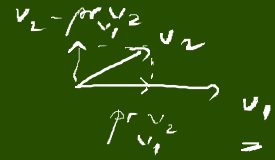
$v \neq 0$

$w = \frac{1}{\|v\|} v$

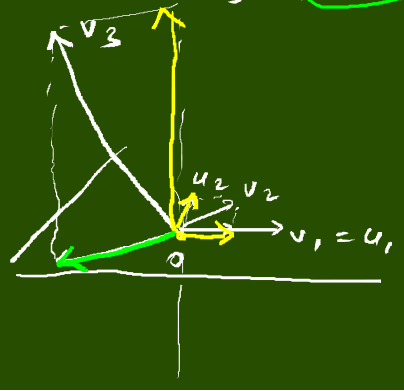
$-w = -\frac{1}{\|v\|} v$

$n = 2$

$v_1 \quad v_2 \Rightarrow \begin{cases} u_1 = v_1 \\ u_2 = v_2 - pr_{u_1} v_2 \end{cases}$



$n = 3$ $\quad u_3 = v_3 - (pr_{u_1} v_3 + pr_{u_2} v_3)$



$v_1 \quad v_2 \quad v_3 \Rightarrow \begin{cases} u_1 = v_1 \\ u_2 = v_2 - pr_{u_1} v_2 \\ u_3 = v_3 - pr_{u_1} v_3 - pr_{u_2} v_3 \end{cases}$

$\mathbb{R}^3$ $\quad v_1 = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$, $v_2 = \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix}$, $v_3 = \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix}$

$\{v_1, v_2, v_3\}$ baza $\subset \mathbb{R}^3$

$\langle v_1, v_2 \rangle = 9 \neq 0 \quad v_1 \not\perp v_2$

$\langle v_1, v_3 \rangle = 9 \neq 0$

$\langle v_2, v_3 \rangle = 27 \neq 0$

$\text{pr}_u v = \frac{\langle v, u \rangle}{\langle u, u \rangle} u$



$\Rightarrow \quad u_1 = v_1 = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$

$u_2 = v_2 - \text{pr}_{u_1} v_2 = \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix} - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ -2 \end{pmatrix}$

$u_3 = v_3 - \text{pr}_{u_1} v_3 - \text{pr}_{u_2} v_3 =$

$= \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix} - \frac{\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} - \frac{\langle v_3, u_2 \rangle}{\langle u_2, u_2 \rangle} \begin{pmatrix} 4 \\ -4 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$

$\Rightarrow \{u_1, u_2, u_3\}$ base ortogonală în $\mathbb{R}^3$

$$q_1 = \frac{1}{\|u_1\|} u_1 = \frac{1}{\sqrt{2^2+1^2+2^2}} u_1 = \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix}$$

$$q_2 = \frac{1}{\|u_2\|} u_2 = \frac{1}{\sqrt{4^2+(-4)^2+(-2)^2}} u_2 = \begin{pmatrix} 2/3 \\ -2/3 \\ -1/3 \end{pmatrix}$$

$$q_3 = \frac{1}{\|u_3\|} u_3 = \frac{1}{\sqrt{(-1)^2+(-2)^2+2^2}} u_3 = \begin{pmatrix} -1/3 \\ -2/3 \\ 2/3 \end{pmatrix}$$

$\Rightarrow B = \{q_1, q_2, q_3\}$ bază ortonormată în $\mathbb{R}^3$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 3/5 \\ 4/5 \\ 0 \end{pmatrix} \times \begin{pmatrix} -4/5 \\ 3/5 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$v = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad [v]_B = ?$$

$$v = \lambda_1 q_1 + \lambda_2 q_2 + \lambda_3 q_3$$

$$= \langle v, q_1 \rangle q_1 + \langle v, q_2 \rangle q_2 + \langle v, q_3 \rangle q_3$$

$$= \frac{2}{3} \cdot q_1 + \frac{2}{3} q_2 + \left(-\frac{1}{3}\right) q_3$$

$Q^t \cdot Q = I_n$ matrice ortogonale

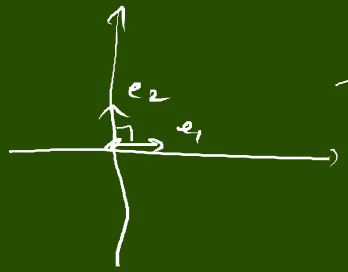
$(n,m)\,(m,n)$

$n = m \Rightarrow Q$ inversabilă, $Q^t = Q^{-1}$

$n = 2$ $\quad Q = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

rotație θ

$Q^t Q = I_2$



Q

